STUDENTS PERFORMANCE DATA ANALYSIS REPORT

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PROJECT OUTLINE :

This report analyzes a student performance dataset to understand patterns in academic results. The dataset includes student scores in mathematics, reading, and writing, along with background factors such as gender, parental education level, and test preparation courses.

PROJECT SUMMARY :

The main goal of this analysis is to identify factors that influence student performance and to understand relationships between different academic subjects. Using Python and data visualization techniques, the dataset was explored to identify trends and patterns.

The analysis shows that reading and writing scores tend to be slightly higher than mathematics scores. Students who completed test preparation courses generally performed better across all subjects. Additionally, some demographic factors such as parental education level appear to have a positive influence on student performance.

These insights can help educational institutions improve learning strategies and provide targeted academic support to students who need it most.

METHODOLOGY :

Before performing the analysis, the dataset was cleaned and prepared to ensure accurate results.

The following steps were performed:

- The dataset was loaded using the Pandas library in Python.
- Column names were standardized for easier analysis.
- The dataset structure was examined using .info() and .describe().
- Missing values were checked using .isnull().sum().
- Any missing or inconsistent values were removed from the dataset.
- Data types were verified to ensure numerical variables could be analyzed correctly.

These steps ensured that the dataset was reliable for analysis.

TOOLS USED :

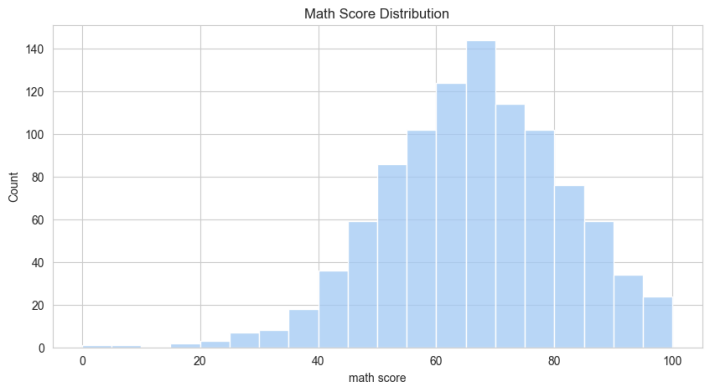
Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook were used for performing data cleaning, analysis, and visualization.

ANALYSIS FINDINGS (KEY ANALYSIS AREAS) :

Several visualizations were created to explore the dataset and understand the distribution of student scores.

- **Distribution of Mathematics Scores :**

  The histogram of mathematics scores shows the spread of student performance in the subject. Most students scored within the mid-range values, indicating moderate performance in mathematics compared to other subjects.

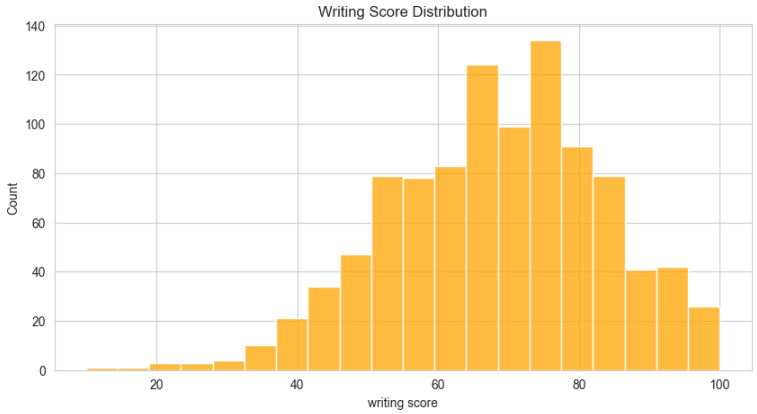


- **Distribution of Reading Scores :**

  Reading scores appear slightly higher overall compared to mathematics. The distribution shows that many students achieved higher marks in reading.

- **Distribution of Writing Scores :**

  Writing scores follow a similar pattern to reading scores, indicating that students tend to perform consistently in language-related subjects.



- **Average Scores by Gender:**

  Bar chart analysis shows differences in average scores between male and female students. In many cases, female students show slightly higher average scores in reading and writing subjects.

STATISTICAL INSIGHTS :

The statistical analysis of the student performance dataset provides insights into the distribution and variation of scores across different subjects.

- The **average scores for reading and writing are slightly higher than mathematics**, indicating stronger performance in language-related subjects.
- The **standard deviation values show moderate variation**, suggesting that student scores are spread across a range but still centered around the mean.
- The **minimum and maximum values** indicate that while some students perform very well, others struggle in certain subjects.
- A **strong positive correlation exists between reading and writing scores**, meaning students who perform well in reading often perform well in writing.
- Mathematics scores show **slightly greater variability**, suggesting that performance in mathematics varies more among students.

These statistical measures help understand the overall academic performance and variability among students.

KEY FINDINGS :

- Students generally perform better in reading and writing compared to mathematics.
- Test preparation courses positively influence student performance.
- Reading and writing scores are strongly correlated.
- Mathematics shows more variability.
- Demographic factors such as parental education level may influence student performance.

BUSINESS RECOMMENDATIONS :

- Strengthen Mathematics Support.
- Encourage Test Preparation Programs.
- Provide Targeted Academic Support.
- Use Data-Driven Decision Making.
- Monitor Student Progress.

CONCLUSION :

This report analyzed a student performance dataset to understand patterns in academic achievement. Through data cleaning, exploratory data analysis, and statistical visualization, several insights were identified. The results show that students generally perform better in reading and writing compared to mathematics. Participation in test preparation courses and certain demographic factors appear to influence academic outcomes. These insights highlight the importance of data-driven decision-making in education. By using performance data effectively, educators can design strategies to improve student learning outcomes and provide targeted support where needed. Future work may include predictive modelling to forecast student performance and identify students who may require early academic intervention.